

Counting possible relations

Will Mayner

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Consider a set S with power set $\mathcal{P}(S)$. Let $n := |S|$ and let $\mathcal{P}_\emptyset(S) := \mathcal{P}(S) \setminus \emptyset$. We wish to count the number of subsets of the disjoint union $P := \mathcal{P}_\emptyset(S) \sqcup \mathcal{P}_\emptyset(S)$ (one copy for causes, one for effects) whose elements have nonempty intersection and have at least 2 elements; that is, we want to find $|R(S)|$, where

$$R(S) := \left\{ r \in \mathcal{P}(P) \quad : \quad \bigcap_{p \in r} p \neq \emptyset \quad \text{and} \quad |r| \geq 2 \right\}$$

Consider $r \in R(S)$ where the intersection of $p \in r$ is a superset of a particular $o \in \mathcal{P}_\emptyset(S)$ (i.e. the purviews in r overlap on at least o). We denote this set by $F(o)$:

$$F(o) := \left\{ r \in R(S) \quad : \quad o \subseteq \bigcap_{p \in r} p \right\}.$$

(This is called the *supertext* of o .)

Lemma 1. *The number of possible relations whose overlap is a superset of o is*

$$|F(o)| = 2^{2^{n-k+1}} - (1 + 2^{n-k+1}),$$

where $k = |o|$, called the relation's degree.

Proof. Let $P' := \mathcal{P}(S \setminus o) \sqcup \mathcal{P}(S \setminus o)$. We can generate all members of $F(o)$ by considering each $r' \in \mathcal{P}(P')$ with 2 or more elements and taking the union of each $p' \in r'$ with o :

$$F(o) = \{ \{p' \cup o : p' \in r'\} \quad : \quad r' \in \mathcal{P}(P') \quad \text{and} \quad |r'| \geq 2 \}$$

Note that $|P'| = 2(2^{n-k}) = 2^{n-k+1}$, and $|\mathcal{P}(P')| = 2^{2^{n-k+1}}$. Since

$$\begin{aligned} |\{r' \in \mathcal{P}(P') : |r'| < 2\}| &= \binom{|\mathcal{P}(P')|}{0} + \binom{|\mathcal{P}(P')|}{1} \\ &= 1 + |\mathcal{P}(P')| \\ &= 1 + 2^{n-k+1}, \end{aligned}$$

we have

$$|F(o)| = 2^{2^{n-k+1}} - (1 + 2^{n-k+1}).$$

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Note that since $|F(o)|$ only depends on k (for fixed n), we can write $f(k) := |F(o)|$.

Theorem 1. *The number of possible relations $|R|$ is*

$$|R| = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \left(2^{2^{n-k+1}} - (1 + 2^{n-k+1}) \right).$$

Proof. Let $O_k := \{o \in \mathcal{P}(S) : |o| = k\}$, so $|O_k| = \binom{n}{k}$. We apply the inclusion-exclusion principle to find $|R|$ using Lemma 1:

$$\begin{aligned} |R| &= \sum_{o \in O_1} |F(o)| - \sum_{o \in O_2} |F(o)| + \sum_{o \in O_3} |F(o)| - \cdots + (-1)^{n-1} \sum_{o \in O_n} |F(o)| \\ &= \sum_{k=1}^n \left((-1)^{k-1} \sum_{o \in O_k} |F(o)| \right) \\ &= \sum_{k=1}^n (-1)^{k-1} |O_k| f(k) \\ &= \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \left(2^{2^{n-k+1}} - (1 + 2^{n-k+1}) \right). \end{aligned}$$

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We can also use Lemma 1 to find the number of possible relations of a given degree k .

Theorem 2. *The number of possible relations $|R_k|$ of degree k is*

$$|R_k| = \binom{n}{k} \left(2^{2^{n-k+1}} - (1 + 2^{n-k+1}) \right) - \binom{n}{k+1} \left(2^{2^{n-k}} - (1 + 2^{n-k}) \right)$$

Proof. Note that

$$|R_k| = \sum_{k=1}^n |R_k| - \sum_{k+1}^n |R_{k+1}|.$$

By Lemma 1, we have

$$\sum_k^n |R_k| = \binom{n}{k} f(o),$$

so

$$\begin{aligned} |R_k| &= \binom{n}{k} f(k) - \binom{n}{k+1} f(k+1) \\ &= \binom{n}{k} \left(2^{2^{n-k+1}} - (1 + 2^{n-k+1}) \right) - \binom{n}{k+1} \left(2^{2^{n-k}} - (1 + 2^{n-k}) \right). \end{aligned}$$

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